How much of summit success is decided before the climb?

Predicting Himalayan summit success from what is known in advance

Dócs Dávid Zoltán Szaniszló Dávid

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Statistical and Machine Learning – WU Vienna

The question

Context

- A Himalayan expedition costs weeks and tens of thousands of euros
- Only 1 in 3 climbers reaches the summit
- Peak, season, team and support are all chosen at home, months before

Why this data set

- Complete record of all expeditions in Nepal, 55,000 climbers
- Non-linear effects and interactions: a fair test for the course methods
- Real pitfalls: leakage and clustered data

Main question: using only what is known before an expedition leaves, how well can we predict whether a climber reaches the summit, and what drives the odds?

1. How well can we predict? 2. Which method is best? 3. What drives the odds? 4. Can we trust it?

Data: The Himalayan Database

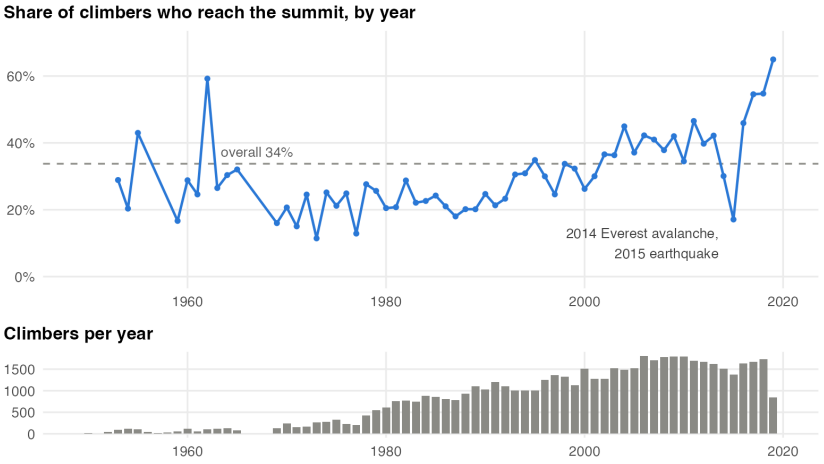
- Elizabeth Hawley's archive of every expedition in the Nepal Himalaya, 1905–2019 (TidyTuesday 2020-09-22)
- Three tables joined to one row per climber: **members** (76,519), **expeditions** (10,364), **peaks** (468)
- **Target:** did the climber reach the summit?

Population: climbers who tried

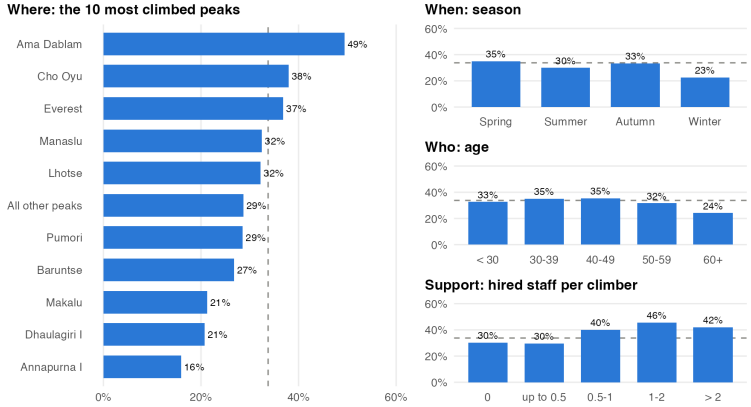
All member records	76,519
1950 or later	76,099
Not hired staff	60,533
Climbing role (no doctors, film crews)	56,443
Expedition attempted the climb	55,062
Peak, sex and season known	**55,049**

55,049 climbers on 9,883 expeditions to 381 peaks. **33.8%** reached the summit. Hired staff are left out: they summit 65% of the time, it is their job.

Exploration: summit rates have doubled since the 1980s



Exploration: where, when, who and how much support



Dashed line: overall summit rate (34%). Peaks differ most (16% to 49%); winter, age 60+ and no hired staff go with low rates.

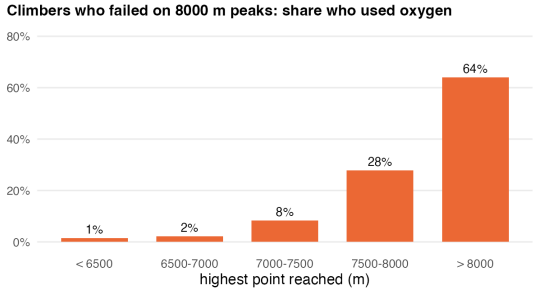
Pre-processing: only what is known before the climb

16 predictors

- *Where:* peak (top 10 + other), height, 8000 m peak, climbed before
- *When:* year, season
- *Support:* hired staff, staff per climber, commercial agency, team size, solo
- *Who:* age (median imputed + missing flag), sex, citizenship (top 15 + other), leader

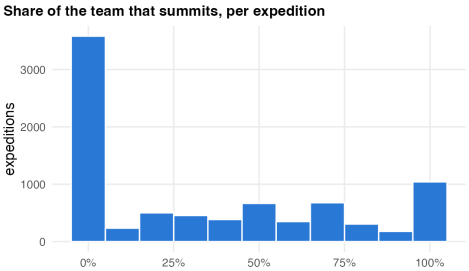
Removed as leakage

highest point reached, death, injury, reason the expedition ended, oxygen use



Oxygen is recorded only once it is used, high on the mountain. It tells us how far the climber got.

Set-up: team-mates share their fate, so we split by expedition



In 43% of expeditions nobody summits, in 12% everybody does.

Splitting

- **Test set:** 20% of expeditions (11,133 climbers), used once at the end
- **Tuning:** 5-fold CV on the training set, folds made of whole expeditions
- Imputation and category grouping learned on the training set

Metrics

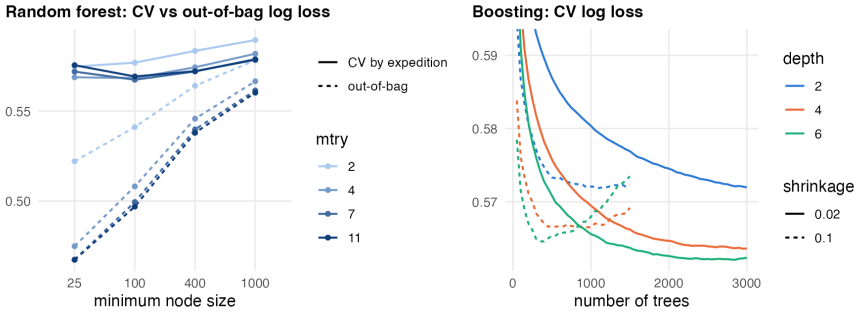
- **Log loss** (main), Brier score, calibration: are the probabilities right?
- **AUC:** are climbers ranked correctly?
- Accuracy says little: "nobody summits" gets 68%

Four methods, from interpretable to flexible

Method	Why	Tuned (grouped CV)	Chosen
Elastic-net logistic regression	Interpretable benchmark; splines for year, age, height; penalty selects variables	mixing α (5 values), penalty λ, 1-SE rule	$\alpha = 0.25$, 38 of 49 terms
Pruned tree	Finds interactions, readable as rules	cost-complexity *cp*, 1-SE rule	30 leaves
Random forest	Averages many decorrelated trees: lower variance	mtry $\times$ minimum node size	mtry 7, node size 100
Gradient boosting	Builds small trees sequentially: lower bias	depth $\times$ shrinkage $\times$ number of trees	depth 6, shrinkage 0.02, 2,850 trees

Each step adds flexibility. If the simple models were enough, the flexible ones would not win on new expeditions.

Tuning: why the out-of-bag error cannot be used here



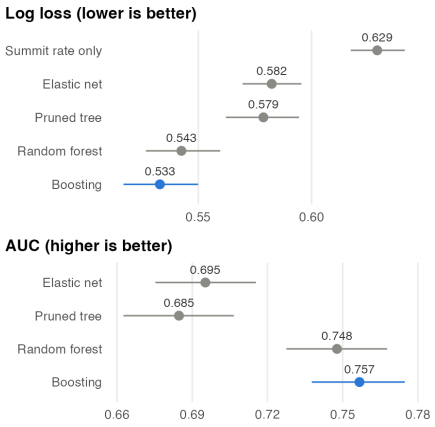
Random forest: out-of-bag error leaves out single *climbers* while their team-mates stay in the bag. It looks much better than CV by expedition (0.50 vs 0.57 log loss) and would pick the smallest nodes, which is worse on new expeditions.

Boosting: with shrinkage 0.1 the error turns up early; with 0.02 deeper trees keep improving and flatten out.

Results on the test set: boosting predicts best

Model	Log loss	Brier	AUC	Acc.
Summit rate only	0.629	0.219	0.500	0.678
Elastic net	0.582	0.198	0.695	0.697
Pruned tree	0.579	0.196	0.685	0.716
Random forest	0.543	0.181	0.748	0.736
Boosting	**0.533**	**0.178**	**0.757**	**0.738**

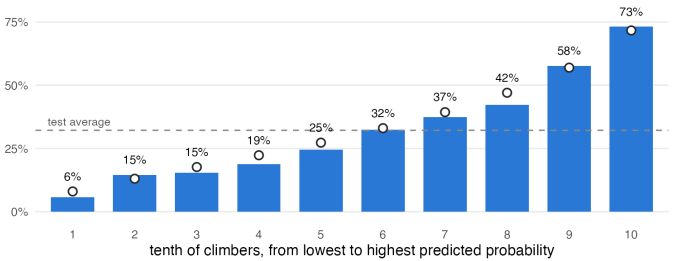
- The flexible models clearly beat the linear model and the single tree: interactions matter
- Boosting beats the forest by 0.010 log loss (95% interval 0.003 to 0.016)
- Lines: 95% intervals from a bootstrap over test expeditions



Answer: pre-climb facts move the odds from 6% to 73%

Observed summit rate of test climbers, by tenth of predicted probability

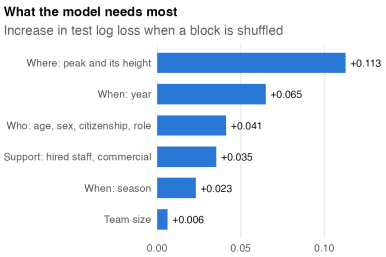
Bars: what happened. Circles: what the boosting model predicted before the climb.

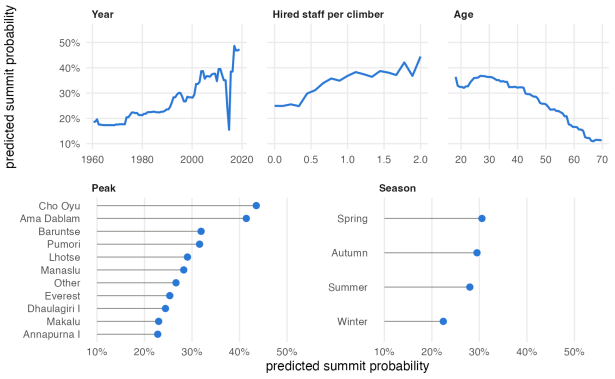


- Without a model every climber gets 34%
- The model separates the test climbers into groups from 6% to 73%
- Predictions match what happened: well calibrated
- Still far from certain: AUC 0.76, log loss 15% below the summit-rate-only baseline

A large part of the outcome is set before the climb starts, but weather, health and luck on the mountain remain.

What drives the odds: the mountain and the era first

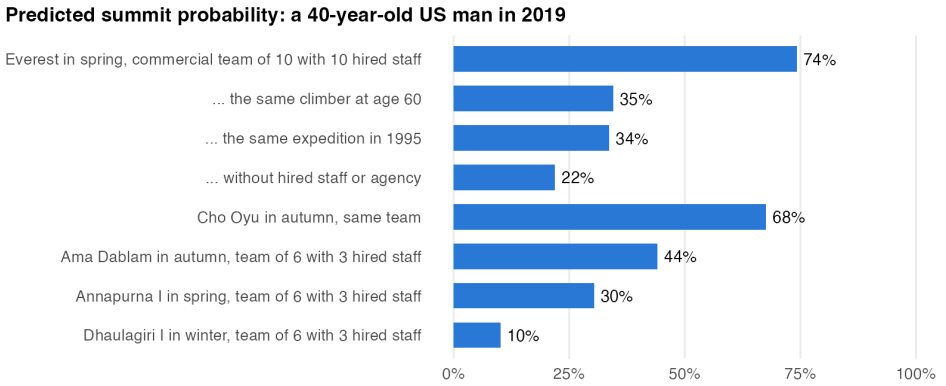




Left: the mountain matters most, then the year; the climber and the support follow, team size hardly matters.

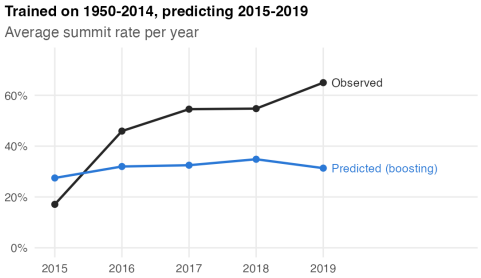
Right (partial dependence): odds rise with the year, dip in 2015 (earthquake), rise with hired staff up to about one per climber, fall after age 40; winter is hardest.

One climber, different plans



The plan changes the predicted odds more than anything else. These are associations, not causal effects: climbers who go without hired staff differ in other ways too.

How far can we trust it? Predicting future seasons



Same settings, trained on 1950–2014, tested on the 7,258 climbers of 2015–2019.

	Log loss	AUC	Predicted
Summit rate only	0.740	0.500	32%
Elastic net	0.703	0.627	34%
Pruned tree	0.668	0.648	42%
Random forest	0.692	0.638	34%
Boosting	0.713	0.661	32%
Observed			*47%*

- The **ranking** still works (AUC 0.63 to 0.66)
- The **level** does not: summit rates jumped after 2015. A year effect can be learned, not extrapolated

Our test set shows how well we explain the past record. For a future season the model must be refitted and recalibrated.

Conclusion

How much of summit success is decided before the climb?
Enough to separate climbers from a 6% to a 73% chance, mostly through the choice of mountain, the era and the support.

Take-aways

1. **Prediction:** boosting reaches AUC 0.76 and is well calibrated on new expeditions
2. **Methods:** boosting $>$ random forest $>$ tree $\approx$ elastic net; flexibility pays because effects interact
3. **Drivers:** where, when, who, how much support
4. **Care:** splitting by expedition and removing oxygen use changed the numbers (leaky model: AUC 0.82)

Limitations

- Weather, health and experience of the climber are not in the data
- Strong changes over time: poor level forecasts for new seasons
- Associations, not causal effects
- Nepal only, up to spring 2019

Thank you!

Questions?

Backup

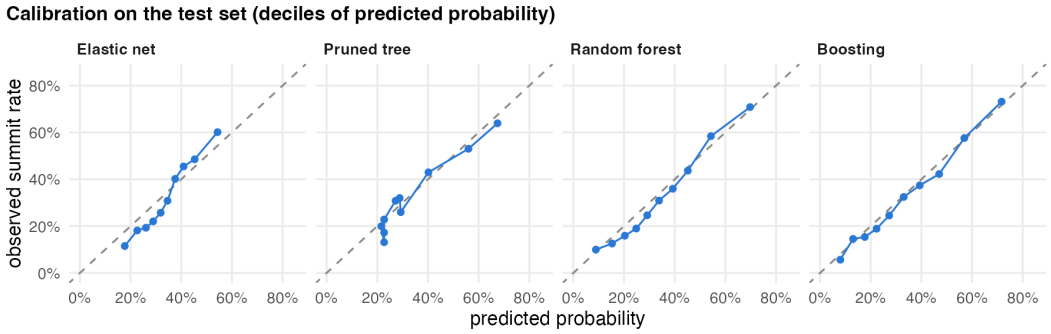
Backup: test results with 95% intervals

Model	Log loss	AUC
Summit rate only	0.629 [0.617, 0.641]	0.500
Elastic net	0.582 [0.570, 0.596]	0.695 [0.675, 0.715]
Pruned tree	0.579 [0.562, 0.594]	0.685 [0.663, 0.707]
Random forest	0.543 [0.527, 0.560]	0.748 [0.728, 0.768]
Boosting	**0.533** [0.517, 0.550]	**0.757** [0.738, 0.775]

Test set: 11,133 climbers from 1,977 expeditions; training set: 43,916 climbers from 7,906 expeditions. Intervals: 1,000 bootstrap samples of whole test expeditions.

Log-loss advantage of boosting: 0.010 over the random forest [0.003, 0.016], 0.046 over the tree [0.035, 0.056], 0.049 over the elastic net [0.039, 0.060]. None of the intervals contains zero.

Backup: calibration of all four models



All models are close to the diagonal. The elastic net and the tree spread their predictions less (20% to 60%); the forest and boosting reach from below 10% to above 70%.

Backup: tuning the elastic net and the tree



Elastic net: CV deviance 1.193 for every $\alpha > 0$ and 1.195 for ridge; $\alpha = 0.25$ with λ from the 1-SE rule (dashed) keeps 38 of 49 terms.

Tree: minimum CV error at 64 leaves; the 1-SE rule (dashed) gives 30 leaves.

Backup: tuning grids

Random forest (300 trees, 5-fold grouped CV)

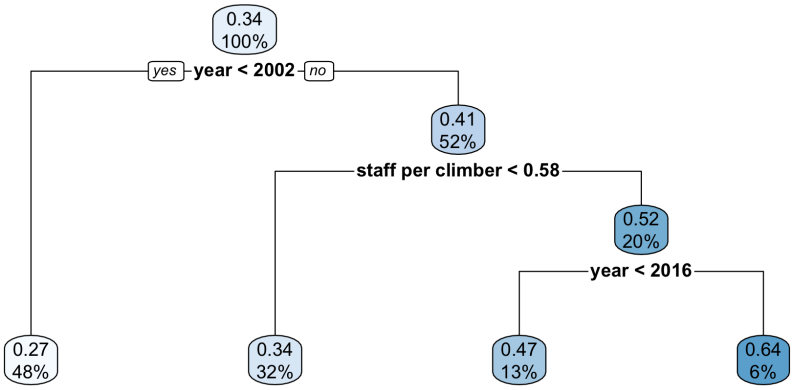
mtry	node size	CV log loss	OOB
7	**100**	**0.567**	0.499
4	100	0.568	0.508
4	25	0.569	0.475
11	100	0.569	0.497
7	25	0.572	0.467
11	25	0.576	0.467
2	1000	0.590	0.578

Grid: mtry 2, 4, 7, 11 $\times$ node size 25, 100, 400, 1000 (selected rows). OOB would pick the smallest nodes.

Boosting (best number of trees per setting)

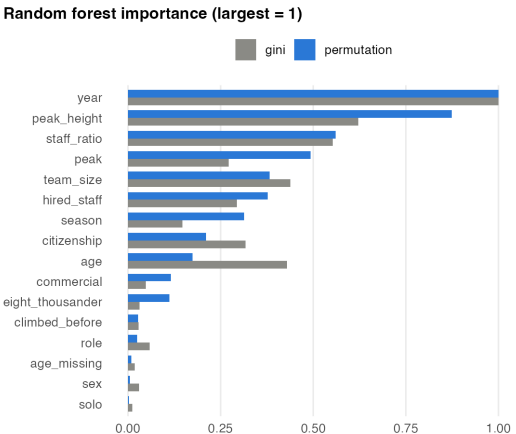
depth	shrinkage	trees	CV log loss
6	**0.02**	**2850**	**0.562**
4	0.02	2950	0.564
6	0.1	350	0.565
4	0.1	1000	0.567
2	0.1	1100	0.572
2	0.02	3000	0.572

Backup: the first splits of the pruned tree



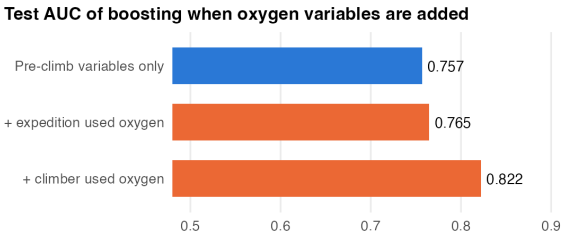
Each box: summit rate and share of training climbers. First split is the era (before 2002: 27%), then hired staff per climber, then the most recent years (2016 and later: 64%).

Backup: importance of single variables



- Permutation and Gini importance agree on the top: year, peak height, staff per climber, peak
- Gini overrates variables with many split points, e.g. age (0.43 vs 0.17)
- Sex, solo and the leader role barely matter
- Boosting, permutation on the test set: year, peak, peak height, staff per climber, season, age, citizenship

Backup: what leakage would have looked like



- Same boosting model plus the oxygen variables
- AUC rises from 0.76 to 0.82, log loss falls from 0.533 to 0.470
- The gain comes from information recorded during the climb, so it would not exist for a real forecast

Backup: data problems and how we handled them

Problem	Handling
Expedition id KANG10101 used twice (1910, 2010)	Join expeditions on id and year
First ascent year "201"	Set to missing
Age missing (about 5%)	Training median + missing flag
213 citizenships, 381 peaks	Top 15 / top 10 by training frequency, rest "other"
525 role names	Keep climbing roles; climber vs leader
Team size and hired staff very skewed	log(1 + x) in the linear model; trees need no transformation
Members who never tried (base camp staff, expeditions that did not start)	Removed from the population